

Mayli Jones

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Personal Statement

Design Engineering MEng student at Imperial College London, entering my third year this October, with hands-on experience developing robotic and electromechanical systems from concept and CAD through to tested physical prototypes. I have strong practical knowledge of 3D printing, design for manufacture and rapid prototyping, and particularly enjoy manufacturing, testing and refining physical components. Curious, adaptable and collaborative, I thrive on unfamiliar challenges and use user feedback to develop practical design improvements.

Education

MEng Design Engineering, Imperial College London 2024 – Present
Year 2: 74% Year 1: 67% London

Awards

1st Place UK, 2nd Overall - (Battery Electric Prototype), Shell Eco-marathon

Competed as part of Imperial College London's Eco-marathon team in the Battery Electric Prototype category, placing 1st in the UK and 2nd overall, and setting a new UK record of 964.0 km/kWh.

Deans List, Imperial - Dyson School of Design Engineering

The Undergraduate Dean's List contains students from the MEng programme that are within the top 10% of their cohort and have achieved the equivalent of a 1st for the year (equal to or above 70% year average).

Best Company Inner East London, Young Enterprise

Served as Managing Director of a Young Enterprise team, leading company strategy, operations and team management, winning the Inner East London Best Company award.

Projects

SwivelCut: Two-Degree-of-Freedom Robotic Cutting Tool, Product Design Engineering 2026

- Co-developed a two-degree-of-freedom robotic cardboard-cutting system using stepper motors, belt-driven transmission and custom mechanical linkages.
- Designed and prototyped CAD components including belt-tensioning mechanisms, spring-loaded suspension systems, modular tool heads and blade assemblies, and produced technical drawings for DFM.
- Integrated and tested mechanical, electronic and control subsystems, refining motor movement, structural stability and cutting performance through iterative prototyping.

Self-Balancing Dancing Segway, Electronics, Signals & Sensors 2026

- Modelled and programmed a self-balancing robot as an inverted-pendulum control system, using PID feedback to maintain stability.
- Developed Python-based beat-detection software to synchronise the robot's movement with music to make it dance to the beat of the song.
- Integrated and tuned the electronic, sensing and control systems for exhibition at the 2026 Design Engineering Exhibition.

MindMeter - Interactive Electromechanical Game, Physical Computing 2025

- Designed and developed an interactive electromechanical game in Fusion 360, creating custom gear trains, mechanisms and sensor-actuated assemblies.
- Used iterative CAD, rapid prototyping and extensive 3D printing to manufacture, test and refine mechanical components and improve reliability.
- Integrated the final mechanical, electronic and control systems for exhibition at the 2026 Design Engineering Exhibition.

- Researched aerodynamic limitations of the roadster archetype, identifying excessive lift as a key issue affecting stability and performance.
- Designed and CAD-modelled a roadster concept intended to mitigate lift, then evaluated performance using CFD analysis in Ansys Fluent, measuring drag and lift coefficients across iterations.
- 3D-printed the final design and tested it in a wind tunnel using a sting mount, comparing physical results against CFD predictions to assess simulation accuracy.

Skills

Engineering & CAD, Onshape, Fusion 360, SolidWorks, AutoCAD, FEA and CFD (Ansys Fluent/Discovery, SolidWorks Simulation), Granta EduPack

Mechanical & Manufacturing, gear trains, belt drives, spring mechanisms, modular assemblies, FDM 3D printing, laser cutting, CNC machining, mechanical assembly

Electronics & Programming, Python, JavaScript, MATLAB, Arduino, ESP32, MicroPython, PID control, sensor and actuator integration, circuit prototyping, motor control

Data Science & ML, regression and classification modelling (SVR, Random Forest, logistic regression), feature engineering, hyperparameter tuning, pandas, scikit-learn

Design Tools, Figma, Adobe InDesign, Canva, Miro, Photoshop, LaTeX

Project Management, Gantt chart planning, stakeholder coordination, cross-functional team delivery

Languages

English: Native

Chinese (Mandarin): Fluent spoken proficiency

Experience

Mechanics Team Member - Shell Eco-marathon, Imperial College London

2025 – Present

Helped design an energy-efficient vehicle for international competition, focused on rear frame design - iterating geometry and material selection using FEA in SolidWorks, and contributing to physical assembly and testing.

Work Experience - Paddington Robotics

2026

Worked alongside engineers on the hands-on assembly and soldering of a robotic supermarket trolley-handling system designed to reduce worker strain and improve efficiency.

GCSE/A-Level Tutor

2023 – Present

One-to-one tuition across maths, physics, chemistry, biology, English, and Mandarin, supporting students from 11+/13+ entrance exams through to GCSE and A-level.

Volunteer - British Heart Foundation

2022 – 2024

Provided customer service, operated the till and supported the day-to-day running of a busy charity shop.

Interests

Music

Grade 8 Piano and Violin, drums, and orchestral performance

Sport

Climbing, ice skating, and swimming, including open water in lakes, rivers and the sea.

Outdoor adventure

Hiking, camping and seeking adventure sports like coasteering and canyoning.